Computers and Fluids 171 (2018) 94–102

Contents lists available at ScienceDirect

Computers and Fluids

journal homepage: www.elsevier.com/locate/compfluid

Decoupling schemes for predicting compressible fluid flows

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ARTICLE INFO

Article history:
Received 8 February 2018
Revised 12 June 2018
Accepted 15 June 2018
Available online 18 June 2018

Keywords:
Compressible fluids
The Euler system
Barotropic fluid
Finite element method
Conservation laws
Two–level schemes
Decoupling scheme

ABSTRACT

In this paper we consider two implicit schemes for the compressible Euler equations, consisting of a scalar advection equation for the density, a vectorial advection equation for the velocity and a barotropic equation of state. The spatial discretization utilizes first order finite elements and the equations are integrated in time by means of the backward Euler scheme, however, the nonlinearity is handled in two different ways. The most straightforward possibility is to solve the nonlinear system by the Newton's method. We show that the resulting discrete fully-nonlinear scheme conserves mass and momentum. Further, we employ a two–level splitting scheme with respect to the physical processes and thus we decouple the mass and momentum conservation equations. The resulting scheme can be applied iteratively at each time level to approximate the solution of the fully nonlinear scheme. The capability of the proposed schemes are illustrated by numerical results for a two–dimensional model problem with an initial density perturbation. The results clearly demonstrate that two iterations at each time level are sufficient to produce non-oscillatory results, without employing any limiters or shock capturing.

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1. Introduction

The models of continuum mechanics (see for example [1,2]) are based on conservation laws for the mass, momentum and energy. In addition, some parameters of the flow, the density for example, have the so called positivity property (monotonicity). It is well known that these important properties of the differential problem of continuum mechanics must be inherited by its discrete counterpart [3,4], i.e., the computed numerical solution should conserve mass, momentum and energy, and if the density is positive, it must remain positive over the entire time interval. Flows of ideal fluids are governed by the Euler equations, whereas the Navier–Stokes equations apply to the flow of viscous fluids. The mathematical aspects of the validation of such models are considered, for example, in the books [5,6]. When discussing the existence of solutions in various Sobolev spaces, the principal problems of the positivity (non–negativity) of the fluid density should be paid a special attention. As mentioned above (see, for instance, [7]) this property must be inherited by the fully discrete counterpart of the mathematical model.

In computational fluid dynamics (CFD), the most important problems are associated with two contradictory requirements. Namely, on one hand it is necessary to construct monotone approximations to the advective terms, and on the other, the conservation laws should be fulfilled at the discrete level. The construction of monotone approximations is discussed in many papers (see, e.g., [8–10]). In [11,12], standard linear approximations for some of the basic problems of continuum mechanics (convection–diffusion problems) are considered. A classical approach in enforcing monotonicity of computed solutions of hyperbolic and convection–dominated problems is by introducing artificial viscosity. For modern artificial viscosity methods we refer the reader to, e.g., [13–16].

For discretization in space, conservative approximations are constructed on the basis of the so-called conservative (divergent) formulations of the equations of continuum mechanics. This approach is very naturally utilized by the integro–interpolation method (balance method) for regular and irregular grids [17], and by the finite volume method [18,19]. Such conservative approximations can also be based on the finite element method [20,21] that is widely used in CFD [22,23].

Most time discretiations of conservation laws utilize explicit schemes that have stability restrictions on the time step. Moreover, in such schemes similar time step restrictions are usually imposed in order to guarantee the monotonicity of the approximate solution. So, it seems natural to try to apply implicit schemes. To solve boundary value problems for partial differential equations, two–level schemes are widely used [9,24,25] (θ–method, schemes with weights). The stability of the two–level schemes in case of linear problems, based on the general theory of stability (well–

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https://doi.org/10.1016/j.compfluid.2018.06.012
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posedness) of operator–difference schemes is presented in [17,26]. In particular, it is possible to formulate sharp (coinciding necessary and sufficient) stability conditions, that are formulated as operator inequalities in finite–dimensional Hilbert spaces.

The current level of research is characterized by the extensive use of explicit schemes in computational fluid dynamics. Approximations in space are associated with a high accuracy of the solution in smooth regions together with an adequate representation of shock waves and other discontinuities. In particular, the Godunov-type MUSCL method, Kolgan–van Leer scheme are in common use [27,28]. The main achievements are related to the construction of various variants of nonlinear approximations of convective transport. This approach is most evidently presented in various modifications of TVD schemes [29,30]. At present, the most interesting are the weighed essentially non-oscillatory (WENO) schemes [31]. We focus on the use of implicit schemes for modeling compressible flows [32,33]. The computational implementation of implicit schemes is much more complicated, but in doing so, we can obtain the unconditional stability and improved properties of the monotonicity for the approximate solution.

In the present work, in Section 2, we consider an initial–boundary value problem for the Euler equations, describing barotropic fluid flows, that represent conservation laws for the mass, momentum, and total mechanical energy. Discretization in space is performed (see Section 3) by standard Lagrange elements for the density and the Cartesian velocity components. To evaluate the approximate solution at the new time level, the fully implicit scheme (the first-order backward Euler scheme) is employed. The resulting approximate solution is mass-conserving and the discrete total mechanical energy satisfies a dissipative estimate. The fully implicit scheme is not convenient for numerical implementation and it is computationally expensive since it requires the solution of a system of coupled nonlinear equations for the density and velocity. In Section 4 we propose a decoupling scheme that belongs to the class of linearized splitting schemes with respect to physical processes [34,35]. The linearization is carried out over the field of advective transport in such a way that at each time level we need to solve individual problems for the density and velocity. This fully decoupled linear scheme can be used iteratively in order to approximate the solution of the fully nonlinear implicit scheme. The capability of the proposed schemes is demonstrated on a model two–dimensional problem with a locally-perturbed initial density and zero initial velocity (see Section 5). Perhaps the most striking feature of the iterative decoupling scheme is that it requires only a small number of iterations (two or three) in order to yield stable and non-oscillatory results. The influence of the grid size in space and time is numerically investigated. It is observed that increasing of the time step results in the monotonization of the numerical solution. The main result of the paper is the proof of the robustness of the linearized decoupling scheme. The scheme involves separate solution of standard advection problems for the density and velocity and demonstrates fast iteration convergence. Such an approach can be applied to other problems of fluid mechanics, e.g., for the numerical solution of initial–boundary value problems for the compressible Navier–Stokes equations. The results of the work are summarized in Section 6.

2. Mathematical model

In this section we consider an initial–boundary value problem for the compressible Euler equations with a barotropic equation of state. The system of equations includes the scalar advection equation for the density and the vector advection equation for the velocity with a given barotropic pressure dependence on the density. The conservation laws for the mass, momentum, and total mechanical energy are discussed below. In our study, we assume that the solution is sufficiently smooth. In this case, a flow of a compressible fluid is governed by the corresponding conservation laws [6,36].

2.1. Barotropic fluid

The continuity equation in a bounded domain Ω has the form

$$\frac{\partial \varrho}{\partial t} + \operatorname{div}(\varrho \boldsymbol{u}) = 0, \quad \boldsymbol{x} \in \Omega, \quad 0 < t \le T, \tag{1}$$

where $\varrho(\boldsymbol{x}, t) > 0$ is the density and $\boldsymbol{u}(\boldsymbol{x}, t)$ is the velocity. The momentum equation can be written in the conservative form

$$\frac{\partial}{\partial t}(\varrho \boldsymbol{u}) + \operatorname{div}(\varrho \boldsymbol{u} \otimes \boldsymbol{u}) + \operatorname{grad} p = 0, \quad \boldsymbol{x} \in \Omega, \quad 0 < t \le T, \tag{2}$$

where $p(\boldsymbol{x}, t)$ is the pressure. The considered fluid is assumed to be barotropic, i.e., we have a known dependence of the pressure on the density $p = p(\varrho)$, $\frac{dp}{d\varrho} > 0$.

Assume that the domain boundaries are rigid and so, the impermeability condition is imposed:

$$(\boldsymbol{u} \cdot \boldsymbol{n}) = 0, \quad \boldsymbol{x} \in \partial\Omega. \tag{3}$$

We also need to specify initial conditions for the density and velocity:

$$\varrho(\boldsymbol{x}, 0) = \varrho^0(\boldsymbol{x}), \quad \boldsymbol{u}(\boldsymbol{x}, 0) = \boldsymbol{u}^0(\boldsymbol{x}), \quad \boldsymbol{x} \in \Omega. \tag{4}$$

The initial–boundary value problem (1)–(4) describes transient flows of an ideal barotropic fluid.

The direct integration of the continuity equation (1) over the domain Ω, taking into account the boundary condition (3), results in the mass conservation law

$$m(t) = m(0), \quad m(t) = \int_\Omega \varrho(\boldsymbol{x}, t) d\boldsymbol{x}. \tag{5}$$

In the Hilbert space $L_2(\Omega)$, we define the scalar product and norm in the standard way:

$$(w, u) = \int_\Omega w(\boldsymbol{x}) u(\boldsymbol{x}) d\boldsymbol{x}, \quad \|w\| = (w, w)^{1/2}.$$

In a similar way, the space of vector functions $\boldsymbol{L}_2(\Omega)$ is defined. If the density is non–negative, the conservation law for the mass can be written as

$$\|\varrho^{1/2}\| = \|\varrho_0^{1/2}\|.$$

This relation can be treated as an a priori estimate for $\varrho^{1/2}$ in $L_2(\Omega)$.

The Eq. (2) directly expresses the conservation law for the momentum. Integrating this equation over Ω, we obtain

$$\int_\Omega \frac{\partial}{\partial t}(\varrho \boldsymbol{u}) d\boldsymbol{x} + \int_{\partial\Omega} p(\varrho) \boldsymbol{n} d\boldsymbol{x} = 0.$$

Thus, we have

$$\boldsymbol{I}(t) = \boldsymbol{I}(0) - \int_{\partial\Omega} p(\varrho) \boldsymbol{n} d\boldsymbol{x}, \quad \boldsymbol{I}(t) = \int_\Omega \varrho \boldsymbol{u} d\boldsymbol{x}. \tag{6}$$

Multiplying by $\boldsymbol{u}$ and taking into account Eq. (1), we can rewrite Eq. (2) as

$$\frac{1}{2}\frac{\partial}{\partial t}(\varrho |\boldsymbol{u}|^2) + \frac{1}{2}\operatorname{div}(\varrho |\boldsymbol{u}|^2 \boldsymbol{u}) + \operatorname{div}(p(\varrho)\boldsymbol{u}) - p \operatorname{div} \boldsymbol{u} = 0.$$

Integration over the domain Ω, in view of (3), leads to

$$\frac{1}{2}\frac{d}{dt}\int_\Omega \varrho |\boldsymbol{u}|^2 \, d\boldsymbol{x} - \int_\Omega p(\varrho) \operatorname{div} \boldsymbol{u} \, d\boldsymbol{x} = 0. \tag{7}$$

The second term in (7) is expressed from the renormalized equation of continuity. Define the pressure potential $\Pi(\varrho)$ from the equation

$$\varrho \frac{d\Pi}{d\varrho} - \Pi(\varrho) = p(\varrho). \tag{8}$$

In particular, for an ideal fluid, we have

$$p(\varrho) = a\varrho^{\gamma}, \quad \Pi(\varrho) = a\frac{\varrho^{\gamma}}{\gamma - 1}, \quad a = \text{const} > 0, \quad \gamma > 1. \tag{9}$$

From the continuity Eq. (1), we have

$$\frac{\partial \Pi}{\partial t} + \operatorname{div}(\Pi \boldsymbol{u}) + p(\varrho)\operatorname{div}\boldsymbol{u} = 0, \quad 0 < t \le T. \tag{10}$$

Integration of the renormalized equation of continuity (10) results in the expression

$$\frac{d}{dt}\int_{\Omega} \Pi\, d\boldsymbol{x} + \int_{\Omega} p(\varrho)\operatorname{div}\boldsymbol{u}\, d\boldsymbol{x} = 0.$$

Adding this equation to (7), we get

$$\frac{d}{dt}\int_{\Omega}\left(\frac{1}{2}\varrho|\boldsymbol{u}|^2 + \Pi(\varrho)\right)d\boldsymbol{x} = 0. \tag{11}$$

Thus, we arrive at the conservation law for the total mechanical energy:

$$E(t) = E(0), \quad E(t) = \int_{\Omega}\left(\frac{1}{2}\varrho|\boldsymbol{u}|^2 + \Pi(\varrho)\right)d\boldsymbol{x}. \tag{12}$$

The Eqs. (5), (6) and (12) are the basic conservation laws for the problem (1)–(4) under the above assumptions on the smoothness of the solution.

2.2. Operator–differential formulation

For the convenience of consideration, we introduce operators of advective (convective) transport for the system of Euler equations. The advection operator $\mathcal{A} = \mathcal{A}(\boldsymbol{u})$ in the divergent form is written as follows:

$$\mathcal{A}(\boldsymbol{u})\varphi = \operatorname{div}(\boldsymbol{u}\varphi). \tag{13}$$

Assuming that the boundary condition (3) is satisfied by the velocity $\boldsymbol{u}$, we obtain

$$(\mathcal{A}(\boldsymbol{u})\varphi, 1) = 0. \tag{14}$$

The continuity Eq. (1) can be written in the form of an operator–differential equation:

$$\frac{d\varrho}{dt} + \mathcal{A}(\boldsymbol{u})\varrho = 0, \quad 0 < t \le T, \tag{15}$$

where the notation $\varrho(t) = \varrho(\boldsymbol{x}, t)$ is used. Similarly, Eq. (2) is written in the form

$$\frac{d}{dt}(\varrho\boldsymbol{u}) + \mathcal{A}(\boldsymbol{u})(\varrho\boldsymbol{u}) + \operatorname{grad} p(\varrho) = 0, \quad 0 < t \le T. \tag{16}$$

For the system of Eqs. (15) and (16) with a prescribed dependence $p(\varrho)$, we consider the Cauchy problem, where the initial conditions (see (4)) have the form

$$\varrho(0) = \varrho^0, \quad \boldsymbol{u}(0) = \boldsymbol{u}^0. \tag{17}$$

For the considered problem (15)–(17), the key point is the property (14) of the advection operator written in divergent form.

3. Implicit two–level scheme

To solve numerically the initial–boundary value problem for the Euler equations, we use the fully implicit (backward Euler) scheme for time stepping together with a finite element discretization in space. The fulfilment of the conservation laws at the discrete level is discussed below.

3.1. Discretization in space

To solve numerically the problem (15)–(17), we employ finite element discretizations in space (see, e.g., [37,38]). For (13), we define the bilinear form

$$a(\varphi, \psi) = \int_{\Omega} \operatorname{div}(\boldsymbol{u}\varphi)\,\psi\, d\boldsymbol{x}.$$

Define the subspace of finite elements $V^h \subset H^1(\Omega)$ and the discrete operator $A = A(\boldsymbol{u})$ as

$$(A(\boldsymbol{u})\varphi, \psi) = a(\varphi, \psi), \quad \forall\, \varphi, \psi \in V^h.$$

Similarly to (14), we have that:

$$(A(\boldsymbol{u})\varphi, 1) = 0. \tag{18}$$

For two– and three–dimensional vector quantities, we use the coordinate–wise representation: $\boldsymbol{u} = (u_1, \dots, u_d)^T$, $d = 2, 3$. For a simple specification of the boundary conditions (3), assume that each part of the boundary of the computational domain is a straight line parallel to one of the coordinate axes. A finite element approximation is used for the individual components of the vector $u_i \in V^h$, $i = 1, \dots, d$.

After discretizing in space, we arrive at a Cauchy problem for a system of semi–discrete operator equations in the corresponding finite–dimensional space, namely, a Cauchy problem for a system of ordinary differential equations. For instance, for (15)–(17), we obtain the problem

$$\frac{d\varrho}{dt} + A(\boldsymbol{u})\varrho = 0, \tag{19}$$

$$\frac{d}{dt}(\varrho\boldsymbol{u}) + A(\boldsymbol{u})(\varrho\boldsymbol{u}) + \operatorname{grad} p(\varrho) = 0, \quad 0 < t \le T. \tag{20}$$

$$\varrho(0) = \varrho_0, \quad \boldsymbol{u}(0) = \boldsymbol{u}_0. \tag{21}$$

Here $\varrho_0 = P\varrho^0$, $\boldsymbol{u}_0 = P\boldsymbol{u}^0$ with P denoting L_2–projection onto V^h.

The solution of the problem (18)–(21) satisfies the same system of conservation laws as the solution of the problem (14)–(17) (see (5), (6), (12)).

3.2. Discretization in time

Let τ be the step of a uniform, for simplicity, grid in time such that $\varphi_n = \varphi(t_n)$, $t_n = n\tau$, $n = 0, 1, \dots, N$, $N\tau = T$. To construct and study time–stepping schemes, the main attention is given to the fulfillment of the corresponding conservation laws (a priori estimates). Such an important problem as the positivity (non–negativity) of the density at each time level requires a more in–depth study and so, it is not considered in the present work.

To solve numerically the problem (19)–(21), the fully implicit scheme is applied. In this case, the approximate solution at the new time level is determined from

$$\frac{\varrho_{n+1} - \varrho_n}{\tau} + A(\boldsymbol{u}_{n+1})\varrho_{n+1} = 0, \tag{22}$$

$$\frac{\varrho_{n+1}\boldsymbol{u}_{n+1} - \varrho_n\boldsymbol{u}_n}{\tau} + A(\boldsymbol{u}_{n+1})(\varrho_{n+1}\boldsymbol{u}_{n+1}) + \operatorname{grad} p(\varrho_{n+1}) = 0,$$
$$n = 0, 1, \dots, N-1, \tag{23}$$

using the prescribed (see (21)) value ϱ_0, $\boldsymbol{u}_0$.

In many applied problems of continuum mechanics, among the basic properties of the approximate solution, one can distinguish properties connected with the fulfillment of the conservation laws for the mass and total energy. The properties of conservatism, multiconservatism, complete conservatism for discrete quantities in computational fluid dynamics are of great importance [39]. To simplify our investigation, assume that the density is positive.

Assumption 1. At each time level $\varrho_n > 0$, $n = 0, 1, \dots, N$.

In view of (18), integration of Eq. (22) over the domain leads to

$$(\varrho_{n+1}, 1) = (\varrho_n, 1), \quad n = 0, 1, \dots, N-1. \tag{24}$$

The equality (24) is a discrete analog the mass conservation law (5). For the momentum conservation law (6), we obtain correspondingly the equality

$$(\varrho_{n+1}\boldsymbol{u}_{n+1}, 1) = (\varrho_n\boldsymbol{u}_n, 1) - \tau \int_{\partial\Omega} p(\varrho_{n+1})\boldsymbol{n}d\boldsymbol{x},$$
$$n = 0, 1, \dots, N-1, \tag{25}$$

which has been obtained by integrating Eq. (23).

An estimate for the total mechanical energy can be established, e.g., following the work [7]. Multiplying Eq. (23) by $\boldsymbol{u}_{n+1}$ and integrating it over Ω, we arrive at

$$\left(\frac{\varrho_{n+1}\boldsymbol{u}_{n+1} - \varrho_n\boldsymbol{u}_n}{\tau}, \boldsymbol{u}_{n+1}\right) + (A(\boldsymbol{u}_{n+1})(\varrho_{n+1}\boldsymbol{u}_{n+1}), \boldsymbol{u}_{n+1})$$
$$+ (\operatorname{grad} p(\varrho_{n+1}), \boldsymbol{u}_{n+1}) = 0. \tag{26}$$

For the first term, we have

$$\begin{aligned}\frac{\varrho_{n+1}\boldsymbol{u}_{n+1} - \varrho_n\boldsymbol{u}_n}{\tau}\boldsymbol{u}_{n+1} &= \frac{1}{2}\frac{\varrho_{n+1}|\boldsymbol{u}_{n+1}|^2 - \varrho_n|\boldsymbol{u}_n|^2}{\tau} \\ &\quad + \frac{1}{2}\frac{\varrho_{n+1}|\boldsymbol{u}_{n+1}|^2 + \varrho_n|\boldsymbol{u}_n|^2 - 2\varrho_n\boldsymbol{u}_n\boldsymbol{u}_{n+1}}{\tau} \\ &= \frac{1}{2}\frac{\varrho_{n+1}|\boldsymbol{u}_{n+1}|^2 - \varrho_n|\boldsymbol{u}_n|^2}{\tau} \\ &\quad + \frac{1}{2}\frac{\varrho_{n+1} - \varrho_n}{\tau}|\boldsymbol{u}_{n+1}|^2 + \frac{1}{2}\varrho_n\frac{|\boldsymbol{u}_{n+1} - \boldsymbol{u}_n|^2}{\tau} \\ &\le \frac{1}{2}\frac{\varrho_{n+1}|\boldsymbol{u}_{n+1}|^2 - \varrho_n|\boldsymbol{u}_n|^2}{\tau} \\ &\quad + \frac{1}{2}\frac{\varrho_{n+1} - \varrho_n}{\tau}|\boldsymbol{u}_{n+1}|^2.\end{aligned}$$

In view of this, from (26), we obtain

$$\frac{1}{2}\left(\frac{\varrho_{n+1}|\boldsymbol{u}_{n+1}|^2 - \varrho_n|\boldsymbol{u}_n|^2}{\tau}, 1\right) + \frac{1}{2}\left(\frac{\varrho_{n+1} - \varrho_n}{\tau}, |\boldsymbol{u}_{n+1}|^2\right)$$
$$+ (A(\boldsymbol{u}_{n+1})(\varrho_{n+1}\boldsymbol{u}_{n+1}), \boldsymbol{u}_{n+1})$$
$$+ (\operatorname{grad} p(\varrho_{n+1}), \boldsymbol{u}_{n+1}) \le 0. \tag{27}$$

From (22) and the definition of the operator A, we have

$$\frac{1}{2}\left(\frac{\varrho_{n+1} - \varrho_n}{\tau}, |\boldsymbol{u}_{n+1}|^2\right) + (A(\boldsymbol{u}_{n+1})(\varrho_{n+1}\boldsymbol{u}_{n+1}), \boldsymbol{u}_{n+1})$$
$$= (A(\boldsymbol{u}_{n+1})(\varrho_{n+1}\boldsymbol{u}_{n+1}), \boldsymbol{u}_{n+1})$$
$$- \frac{1}{2}(A(\boldsymbol{u}_{n+1})\varrho_{n+1}), |\boldsymbol{u}_{n+1}|^2) = 0.$$

This makes possible to rewrite the inequality (27) as

$$\frac{1}{2}\left(\frac{\varrho_{n+1}|\boldsymbol{u}_{n+1}|^2 - \varrho_n|\boldsymbol{u}_n|^2}{\tau}, 1\right) - (p(\varrho_{n+1}), \operatorname{div}\boldsymbol{u}_{n+1}) \le 0. \tag{28}$$

To estimate the second term on the left–hand side of the inequality (28), we apply the discrete analogue of the renormalized equation of continuity (10). Multiply the continuity Eq. (22) by $\frac{d\Pi}{d\varrho}(\varrho_{n+1})$:

$$\frac{\varrho_{n+1} - \varrho_n}{\tau}\frac{d\Pi}{d\varrho}(\varrho_{n+1}) + A(\boldsymbol{u}_{n+1})\varrho_{n+1}\frac{d\Pi}{d\varrho}(\varrho_{n+1}) = 0. \tag{29}$$

The following equality takes place:

$$\Pi(\varrho_{n+1}) - \Pi(\varrho_n) = \frac{d\Pi}{d\varrho}(\varrho_{n+1})(\varrho_{n+1} - \varrho_n)$$
$$- \frac{1}{2}\frac{d^2\Pi}{d\varrho^2}(\widetilde{\varrho}^{n+1})(\varrho_{n+1})(\varrho_{n+1} - \varrho_n)^2,$$

where

$$\widetilde{\varrho}^{n+1} \in [\min(\varrho_n, \varrho_{n+1}), \ \max(\varrho_n, \varrho_{n+1})].$$

Assumption 2. Assume that

$$\frac{d^2\Pi}{d\varrho^2}(\varrho) \ge 0, \tag{30}$$

Under these natural assumptions, we get

$$\frac{d\Pi}{d\varrho}(\varrho_{n+1})(\varrho_{n+1} - \varrho_n) \ge \Pi(\varrho_{n+1}) - \Pi(\varrho_n).$$

For the second term in (29), taking into account (8), we have

$$A(\boldsymbol{u}_{n+1})\varrho_{n+1}\frac{d\Pi}{d\varrho}(\varrho_{n+1})$$
$$= \varrho_{n+1}\frac{d\Pi}{d\varrho}(\varrho_{n+1})\operatorname{div}\boldsymbol{u}_{n+1} + \boldsymbol{u}_{n+1}\operatorname{grad}\Pi(\varrho^{n+1})$$
$$= \operatorname{div}(\Pi(\varrho_{n+1})\boldsymbol{u}_{n+1}) + p(\varrho_{n+1})\operatorname{div}\boldsymbol{u}_{n+1}.$$

In view of this, integration of Eq. (29) results in

$$\left(\frac{\Pi(\varrho_{n+1}) - \Pi(\varrho_n)}{\tau}, 1\right) + (p(\varrho_{n+1}), \operatorname{div}\boldsymbol{u}_{n+1}) \le 0. \tag{31}$$

Combining (28) and (31), we obtain the inequality

$$\left(\frac{1}{2}\varrho_{n+1}|\boldsymbol{u}_{n+1}|^2 + \Pi(\varrho_{n+1}), 1\right) \le \left(\frac{1}{2}\varrho_n|\boldsymbol{u}_n|^2 + \Pi(\varrho_n), 1\right). \tag{32}$$

Comparing (32) with (12), we can conclude that at the discrete level, instead of fulfillment of the conservation law for the total energy, we observe a decrease of the energy. The conservation of the total energy takes place for smooth solutions of the problem under the consideration. For problems with discontinuous solutions, dissipation of the total energy (an inequality of type (32)) is natural. It should be noted that property (32) has been established under the additional assumption (30). The result of our consideration can be expressed in the following statement.

Proposition 1. *The fully implicit scheme (22) and (23) produces an approximate solution of the problem (19)–(21) that satisfies the mass conservation law in the form (24) and the momentum conservation law (25). Moreover, if the Assumptions 1 and 2 hold, the estimate (32) for the total mechanical energy is also fulfilled.*

4. Decoupling schemes

In order to decouple the nonlinear scheme (22) and (23) a linearized scheme is used, where the solution at the new time level is evaluated taking the advection velocity from the previous time level. Using such a linearization, an iterative process is constructed for the numerical approximation of the fully implicit scheme. The approximate solution at the new time level is produced by sequentially solving, first, a linear advection problem for the density and, secondly, a linear problem for the velocity.

4.1. Linearized scheme

We focus on the use of such time–stepping techniques that possess the following properties:

- the transition to a new time level is performed by solving linear problems;
- splitting with respect to physical processes is employed, namely, the problems for the density and velocity are solved separately, the problems for the velocity components being also decoupled.

An example of the simplest decoupling scheme for the Euler equations system (19) and (20) is the linearized scheme, where the advective transport involves the velocity from the previous time level.

Instead of (22) and (23), we employ the scheme

$$\frac{\varrho_{n+1}-\varrho_n}{\tau}+A(\boldsymbol{u}_n)\varrho_{n+1}=0, \tag{33}$$

$$\frac{\varrho_{n+1}\boldsymbol{u}_{n+1}-\varrho_n\boldsymbol{u}_n}{\tau}+A(\boldsymbol{u}_n)(\varrho_{n+1}\boldsymbol{u}_{n+1})+\operatorname{grad} p(\varrho_{n+1})=0,$$
$$n=0,1,\dots,N-1. \tag{34}$$

First, from the linear transport Eq. (33), we evaluate the density at the new time level. Next, from the linear decoupled system (34) for the velocity components, we calculate the velocity $\boldsymbol{u}_{n+1}$.

Remark 1. The system of Eqs. (34) with a given density is, in the general case, coupled for individual cartesian velocity components. In the case, where all parts of the boundary of the computational domain $\partial\Omega$ are parallel to the axes of the Cartesian coordinate system, the system of Eqs. (34) is decoupled and so, we can evaluate independently the individual components of the velocity.

For the linearized scheme, the discrete analogs of the mass conservation law (see (24)) and the momentum conservation law (see (25)) hold.

4.2. Iterative decoupling scheme

On the basis of the linearized scheme (33) and (34), it is possible to construct an iterative algorithm for the numerical implementation of the fully implicit scheme (22) and (23). The approximate solution for ϱ_{n+1}, $\boldsymbol{u}_{n+1}$ at the k–th iteration is denoted by ϱ^k_{n+1}, $\boldsymbol{u}^k_{n+1}$, with the initial approximation given by the previous time level solution:

$$\varrho^0_{n+1}=\varrho_n,\quad \boldsymbol{u}^0_{n+1}=\boldsymbol{u}_n. \tag{35}$$

Assume that the new approximation at the new time level is calculated, when the previous K iterations have been done. Similarly to (33) and (34), we use the system of equations:

$$\frac{\varrho^{k+1}_{n+1}-\varrho_n}{\tau}+A(\boldsymbol{u}^k_{n+1})\varrho^{k+1}_{n+1}=0, \tag{36}$$

$$\frac{\varrho^{k+1}_{n+1}\boldsymbol{u}^{k+1}_{n+1}-\varrho_n\boldsymbol{u}_n}{\tau}+A(\boldsymbol{u}^k_{n+1})(\varrho^{k+1}_{n+1}\boldsymbol{u}^{k+1}_{n+1})+\operatorname{grad} p(\varrho^{k+1}_{n+1})=0,$$
$$k=0,1,\dots,K-1, \tag{37}$$

where

$$\varrho_{n+1}=\varrho^K_{n+1},\quad \boldsymbol{u}_{n+1}=\boldsymbol{u}^K_{n+1},\quad n=0,1,\dots,N-1. \tag{38}$$

Thus, at each iteration, we first solve the linear problem for the density and then compute the solution of the linear problem for the velocity.

5. Numerical results

In this section the capability of the fully implicit and the decoupling schemes are illustrated by numerical results for a model two–dimensional problem with a density perturbation of an initially resting fluid.

5.1. Test problem

The problem (1)–(4) is considered in the square

$$\Omega=\{\boldsymbol{x}\mid \boldsymbol{x}=(x_1,x_2),\quad -5<x_1<5,\quad -5<x_2<5\}.$$

Table 1
Convergence of Newton's method.

iteration	$\tau=0.01$	$\tau=0.005$	$\tau=0.0025$
1	2.219e-03	5.602e-04	1.404e-04
2	5.104e-08	9.100e-10	1.458e-11
3	5.566e-15	9.994e-15	1.788e-14

Assume that the dependence of the density on the pressure has the form ((9) with $a=1$, $\gamma=1.4$. We simulate the motion of the initially resting fluid ($\boldsymbol{u}^0(\boldsymbol{x})=0$ in (4)) with the initial density (see (4)) specified in the form

$$\varrho^0(\boldsymbol{x})=1+\alpha\exp(-\beta|\boldsymbol{x}|^2),$$

where $\alpha=2$ and $\beta=20$.

5.2. Fully implicit scheme

The problem is solved using a uniform triangulation of Ω consisting of M segments in each direction. Piecewise linear P_1 finite elements are employed for discretization in space. To implement the fully implicit scheme (22) and (23) for the nonlinear discrete problem at the new time level, the Newton method with a direct solver for the corresponding system of linear algebraic equations is applied.

The time evolution of the compressible fluid flow is shown in Fig. 1, which presents the density at various time moments. In this calculation, we used a spatial grid with $M=200$, the time step was chosen to be $\tau=0.005$. Fig. 2 demonstrates dynamics of the flow. Time-histories of the density at the center of the computational domain (ϱ_0), the maximum ($\varrho_{\max}$) and minimum ($\varrho_{\min}$) values of the density over the entire domain Ω are given. We observe quick local changes in the density at the beginning of the process. Such peculiarities of the solution of the dynamic problem can be taken into account, in particular, by means of an adaptive choice of the time step.

The iterative method for solving the discrete problem at each new time moment converges very quickly (two or three iterations are enough). Table 1 demonstrates convergence of the iterative process for the first step in time. Here, we present the relative error for the first three iterations for the model problem obtained with $M=200$ and various time steps.

The accuracy of the approximate solution of the test problem is further illustrated by the data on the density ion the line $x_2=0$. The solution calculated on ta grid with $M=50$ using various time steps is shown in Fig. 3. Similar data for $M=100$ and $M=200$ are shown in Figs. 4 and 5, respectively. It is clear that the accuracy of the approximation of the leading edge of the wave, improves fast with the refinement of the spatial grid. Also, we observe the effect of smoothing, namely, elimination of non–monotonicity with the increase of the time step.

The considered implicit schemes belong to the class of unconditionally stable ones. This is their most important advantage over explicit schemes, which are unstable without using some stabilization procedures even in numerical solving elementary linear problems of advection [40]. Therefore, a choice of time step is subject to requirements of accuracy and monotonicity of the approximate solution. These values of the time step can be compared with the Courant number, with the Courant–Friedrichs–Lewy (CFL) stability condition for classical explicit schemes with directional differences, schemes with an artificial diffusion.

For the considered test problem, the Courant number over the calculation domain at the time moment t is determined by the expression

$$C(t)=\Big(\max_{\boldsymbol{x}\in\Omega}|u_1|+\max_{\boldsymbol{x}\in\Omega}|u_2|\Big)\tau\frac{10}{M},\quad \boldsymbol{u}=(u_1,u_2).$$

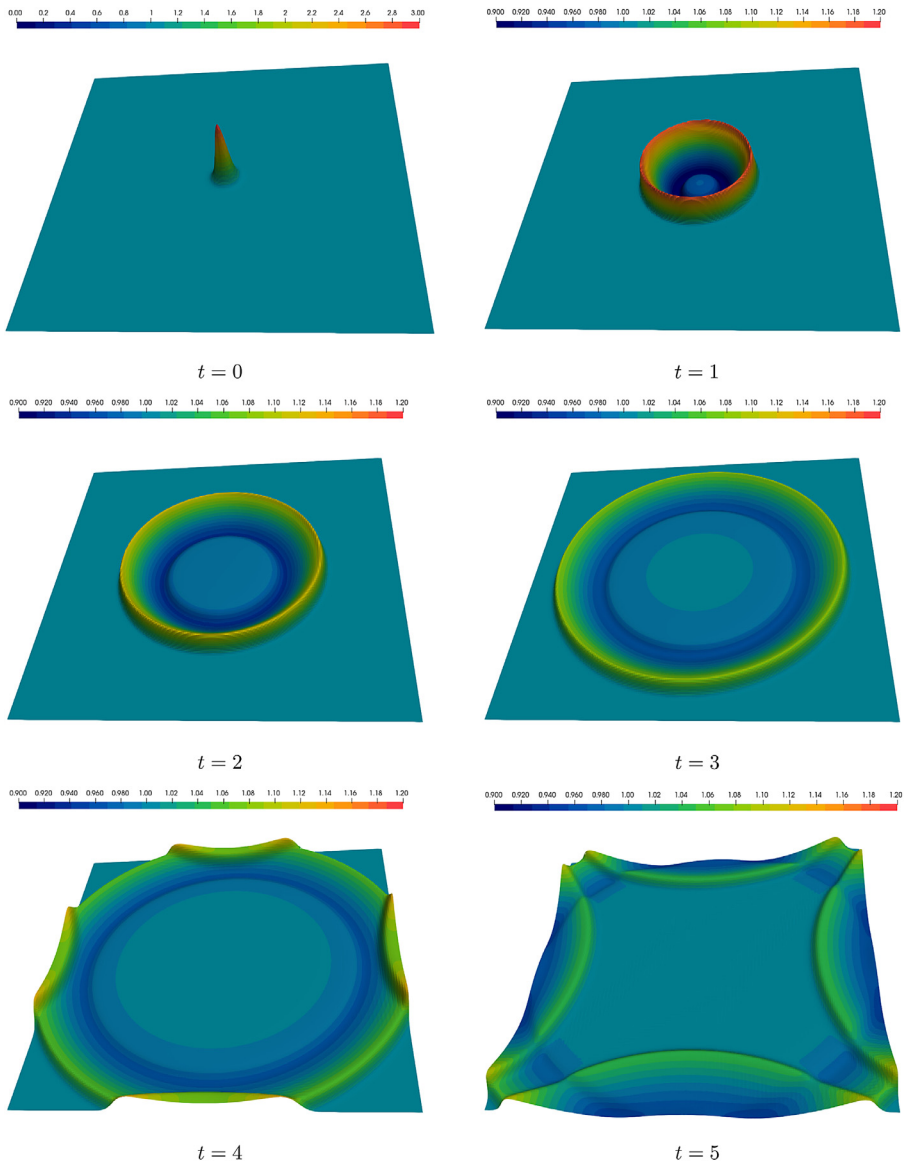

Fig. 1. The density at various time moments.

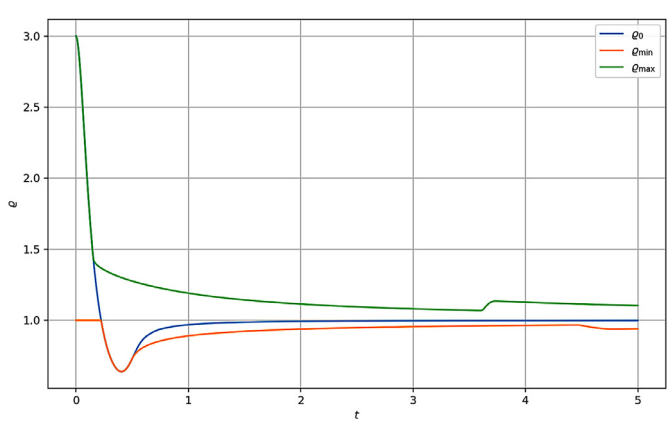

Fig. 2. Time–histories of the density (central, maximal and minimal values).

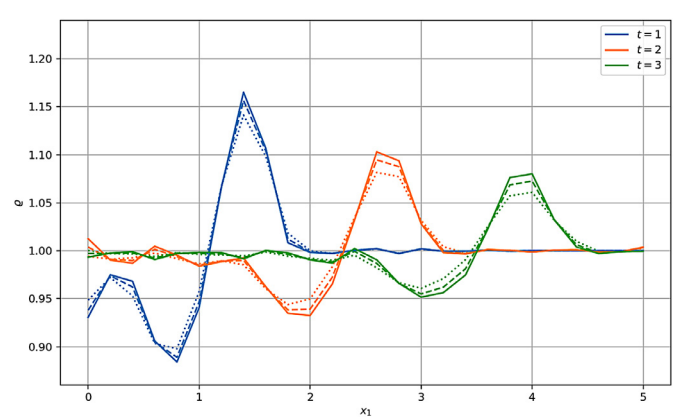

Fig. 3. The solution of the problem at various time moments calculated on the grid $M = 50$: dotted line – $\tau = 0.01$, dashed – $\tau = 0.005$, solid – $\tau = 0.0025$.

The Courant number for different grids in space and the time step $\tau = 0.01$ is shown in Fig. 6. At the developing stage, the Courant number decreases with time, i.e., perturbations propagate with a lower speed. Here, the calculations for the test problem are carried out with small Courant numbers. Once again, we emphasize that the time step is not limited by stability. Moreover, an increase in the time step leads to a stronger smoothing of the approximate solution.

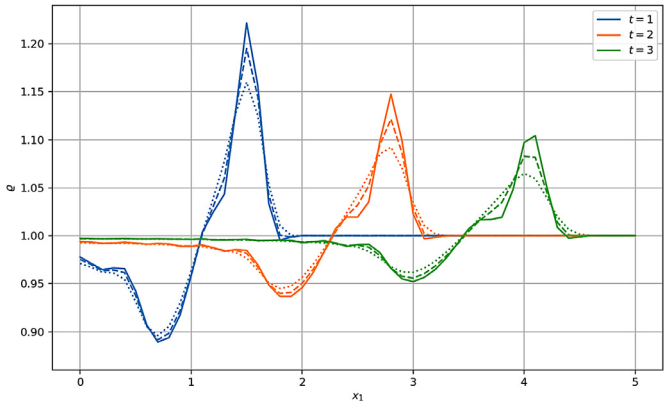

Fig. 4. The solution of the problem at various time moments calculated on the grid $M = 100$: dotted line – $\tau = 0.01$, dashed – $\tau = 0.005$, solid – $\tau = 0.0025$.

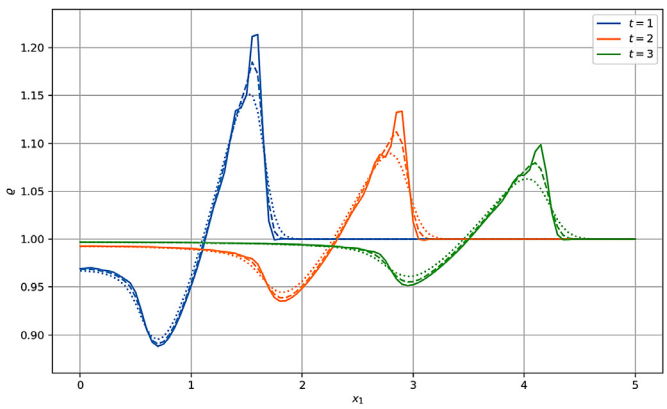

Fig. 5. The solution of the problem at various time moments calculated on the grid $M = 200$: dotted line – $\tau = 0.01$, dashed – $\tau = 0.005$, solid – $\tau = 0.0025$.

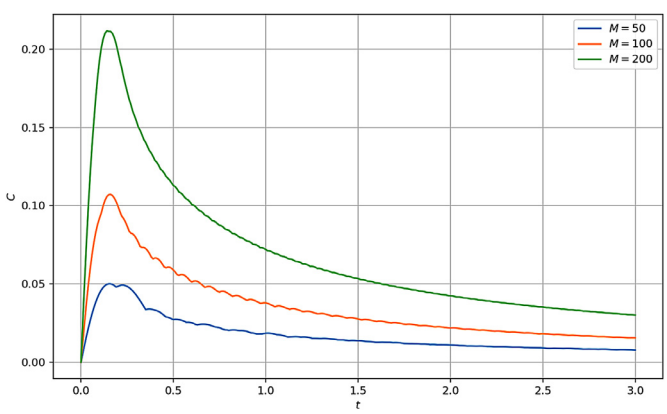

Fig. 6. Time-evolution of the Courant number calculated with $\tau = 0.01$.

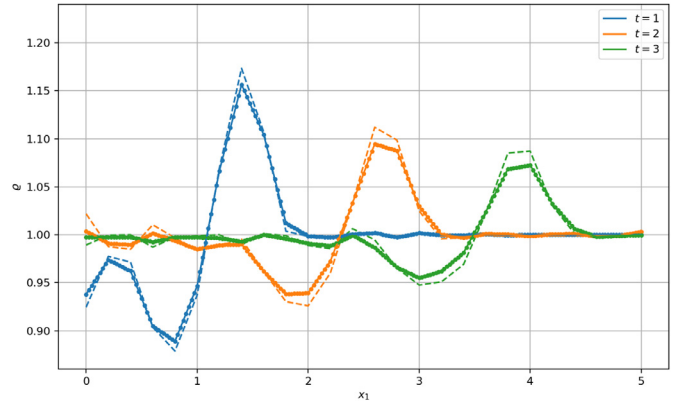

Fig. 7. The solution of the problem at various time moments calculated on the grid $M = 50$: dashed line – $K = 1$, dotted – $K = 2$, solid – $K = 5$.

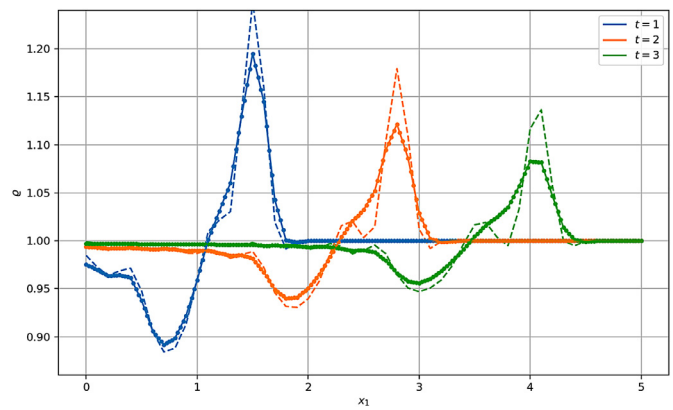

Fig. 8. The solution of the problem at various time moments calculated on the grid $M = 100$: dashed line – $K = 1$, dotted – $K = 2$, solid – $K = 5$.

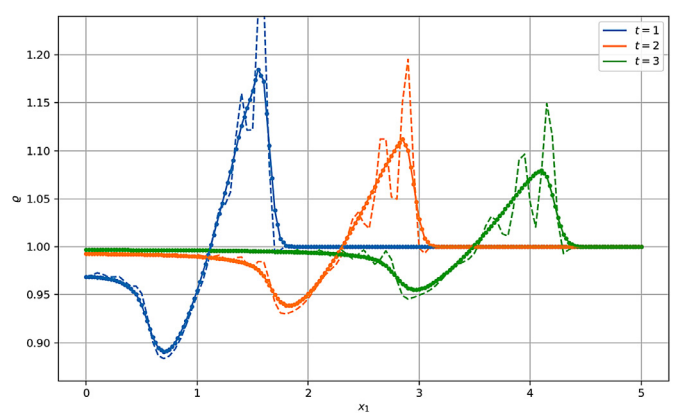

Fig. 9. The solution of the problem at various time moments calculated on the grid$M = 200$: dashed line – $K = 1$, dotted – $K = 2$, solid – $K = 5$.

5.3. Decoupling scheme

In using the decoupling scheme (35)–(38), the greatest interest is related to the convergence rate of the iterative process. The time step in the calculations was equal to $\tau = 0.005$. We present numerical results for the model problem under consideration obtained, on different grids in space. The dependence of the solution on the number of iterations for the grid with $M = 50$ is shown in Fig. 7. Figs. 8 and 9 presents similar data for grids with $M = 100$ and $M = 200$, respectively. It is easy to see see that on the finest grid (see Fig. 9) the linearized scheme (34) and (34) ($K = 1$ in (35)–(38)) yields a substantially non–monotonic solution, which is monotonized on subsequent iterations.

The main conclusion of our study is the demonstration of high computational efficiency of the iterative decoupling scheme (35)–(38). Namely, for the problems under consideration it is sufficient to do only two iterations using (35)–(38).

For the above time–stepping methods, the key point is a violation of the conservation law for the total energy. For the fully implicit scheme (22) and (23), instead of conservation of the energy (see the estimate (32)), decreasing of the total energy is observed.

The dynamics of the total mechanical energy using the linearized scheme (33) and (34) ($K = 1$) and iterative decoupling schemes (35)–(38)) for $K = 5$ on various grids is shown in Figs. 10–12. Here, according to (12), $E(t_n)$ is calculated at each time

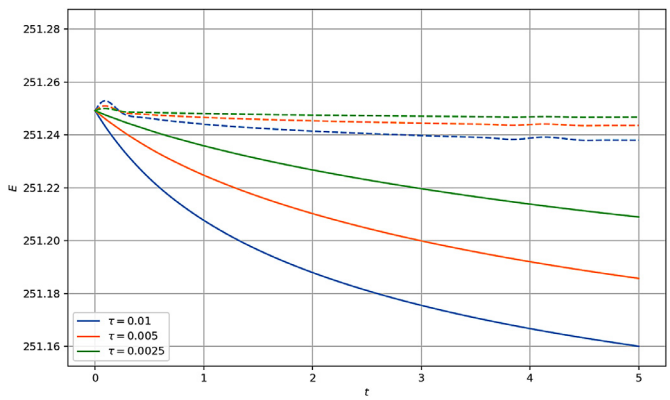

Fig. 10. Time–history of the total mechanical energy for various time steps obtained on the grid with $M = 50$: dashed line – $K = 1$, solid – $K = 5$.

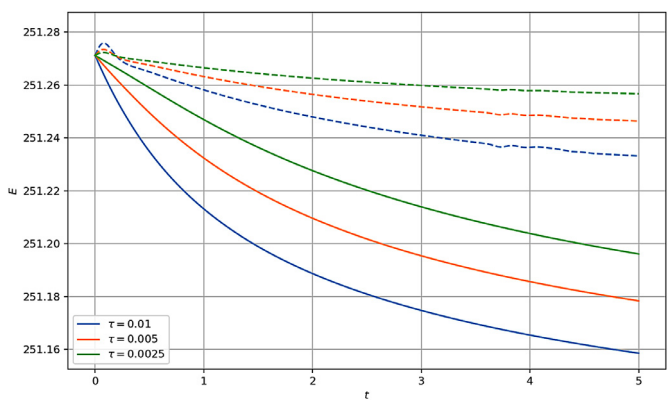

Fig. 11. Time–history of the total mechanical energy for various time steps obtained on the grid with $M = 100$: dashed line – $K = 1$, solid – $K = 5$.

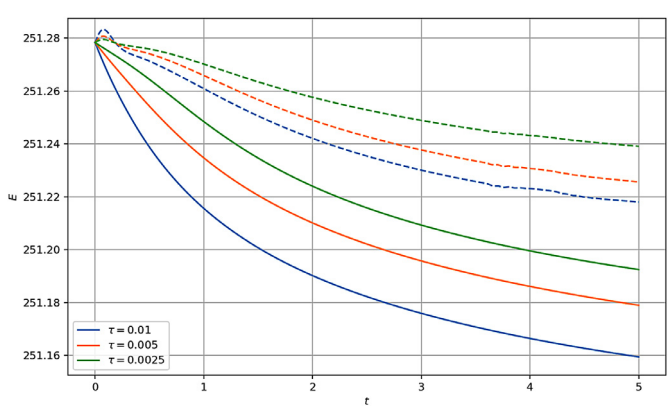

Fig. 12. Time–history of the total mechanical energy for various time steps obtained on the grid with $M = 200$: dashed line – $K = 1$, solid – $K = 5$.

moment as follows:

$$E(t_n) = \int_\Omega \left(\frac{1}{2}\varrho_n|\boldsymbol{u}_n|^2 + \Pi(\varrho_n)\right)d\boldsymbol{x}, \quad n = 0, 1, \ldots, N.$$

For $K = 5$, the energy of the solution obtained using the decoupling scheme (35)–(38) practically coincides with the one for the solution of the fully implicit scheme (22) and (23). This indicates that the conservation law for the total energy is satisfied with a good accuracy. Decreasing of the time step results in increasing of the accuracy of the conservation law fulfillment.

6. Conclusions

- For predicting flows of an ideal barotropic liquid, the standard finite-element approximation in space is used along with the fully implicit two-level discretization in time (backward Euler). An approximate solution of the nonlinear problem at a new time level is evaluated using the iterative Newton method. Peculiarities of the conservation of the mass, momentum, and total mechanical energy are investigated.
- An iterative decoupling scheme is proposed for dynamic problems of barotropic fluid mechanics, where the approximate solution at the new time level is obtained by solving linear problems for the density and velocity.
- The performance of the fully implicit scheme in suppressing unphysical oscillations is demonstrated by calculations for a model two-dimensional problem. Such a monotonization of the approximate solution can be combined with other known approaches. In particular, there are good prospects for using different variants of linear and nonlinear higher-order approximations in space.

Acknowledgments

The publication has been prepared with the support of the RUDN University Program 5–100.

References

[1] Batchelor GK. An introduction to fluid dynamics. Cambridge University Press; 2000.
[2] Landau LD, Lifshitz E. Fluid mechanics. Butterworth-Heinemann; 1987.
[3] Anderson J. Computational fluid dynamics: the basics with applications. McGraw-Hill; 1995.
[4] Wesseling P. Principles of computational fluid dynamics. Springer; 2010.
[5] Lions P-L. Mathematical topics in fluid mechanics: incompressible models. Oxford University Press; 1996.
[6] Lions P-L. Mathematical topics in fluid mechanics: compressible models. Oxford University Press; 1998.
[7] Feireisl E, Karper TG, Pokorný M. Mathematical theory of compressible viscous fluids: analysis and numerics. Springer; 2016.
[8] Kulikovskii AG, Pogorelov NV, Semenov AY. Mathematical aspects of numerical solution of hyperbolic systems. Taylor & Francis; 2000.
[9] Hundsdorfer WH, Verwer JG. Numerical solution of time-dependent advection-diffusion-reaction equations. Springer Verlag; 2003.
[10] Kuzmin D. A guide to numerical methods for transport equations. University Erlangen–Nuremberg; 2010.
[11] Morton KW, Kellogg RB. Numerical solution of convection-diffusion problems. Chapman & Hall London; 1996.
[12] Samarskii AA, Vabishchevich PN. Numerical methods for solving convection-diffusion problems. Moscow: URSS; 1999. In Russian
[13] Guermond J-L, Pasquetti R, Popov B. Entropy viscosity method for nonlinear conservation laws. J Comput Phys 2011;230(11):4248–67.
[14] Hartmann R, Houston P. Adaptive discontinuous galerkin finite element methods for nonlinear hyperbolic conservation laws. SIAM J Sci Comput 2002;24(3):979–1004.
[15] Kurganov A, Liu Y. New adaptive artificial viscosity method for hyperbolic systems of conservation laws. J Comput Phys 2012;231:8114–32.
[16] Guermond J-L, Popov B, Tomov V. Entropy–viscosity method for the single material euler equations in lagrangian frame. Comput Methods Appl Mech Eng 2016;300:402–26.
[17] Samarskii AA. The theory of difference schemes. New York: Marcel Dekker; 2001.
[18] LeVeque RJ. Finite volume methods for hyperbolic problems. Cambridge University Press; 2002.
[19] Versteeg H, Malalasekra W. An introduction to computational fluid dynamics: the finite volume method. Prentice Hall; 2007.
[20] Ern A, Guermond J-L. Theory and practice of finite elements. Springer; 2004.
[21] Larson MG, Bengzon F. The finite element method: theory, implementation and applications. Springer; 2013.
[22] Donea J, Huerta A. Finite element methods for flow problems. Wiley; 2003.
[23] Zienkiewicz OC, Taylor RL, P N. The finite element method for fluid dynamics. Butterworth-Heinemann; 2013.
[24] Ascher UM. Numerical methods for evolutionary differential equations. Soc Ind Appl Math; 2008.
[25] LeVeque RJ. Finite difference methods for ordinary and partial differential equations: steady-state and time-dependent problems. Soc Ind Appl Math; 2007.

[26] Samarskii AA, Matus PP, Vabishchevich PN. Difference schemes with operator factors. Dordrecht: Kluwer Academic; 2002.
[27] Kolgan VP. Application of the principle of minimizing the derivative to the construction of finite-difference schemes for computing discontinuous solutions of gas dynamics. J Comput Phys 2011;230(7):2384–90.
[28] Van Leer B. Towards the ultimate conservative difference scheme. V. A second-order sequel to Godunov's method. J Comput Phys 1979;32(1):101–36.
[29] Harten A. High resolution schemes for hyperbolic conservation laws. J Comput Phys 1983;49(3):357–93.
[30] LeVeque RJ. Numerical methods for conservation laws. Berlin: Birkhauser; 1992.
[31] Shu C-W. Essentially non-oscillatory and weighted essentially non-oscillatory schemes for hyperbolic conservation laws. In: Advanced numerical approximation of nonlinear hyperbolic equations. Springer; 1998. p. 325–432.
[32] Beam RM, Warming RF. An implicit finite-difference algorithm for hyperbolic systems in conservation-law form. J Comput Phys 1976;22(1):87–110.
[33] Godlewski E, Raviart P-A. Numerical approximation of hyperbolic systems of conservation laws. Springer; 2013.
[34] Marchuk GI. Splitting and alternating direction methods. In: Ciarlet PG, Lions J-L, editors. Handbook of numerical analysis, I. North-Holland; 1990. p. 197–462.
[35] Vabishchevich PN. Additive operator-difference schemes: splitting schemes. Berlin: de Gruyter; 2014.
[36] Novotny A, Straskraba I. Introduction to the mathematical theory of compressible flow, 27. Oxford University Press on Demand; 2004.
[37] Thomée V. Galerkin finite element methods for parabolic problems. Berlin: Springer Verlag; 2006.
[38] Brenner SC, Scott LR. The mathematical theory of finite element methods. New York: Springer; 2008.
[39] Samarskii AA, Popov IP. Finite-difference schemes of gasdynamics. URSS; 2004. In Russian
[40] Vabishchevich PN. Two-level schemes for the advection equation. J Comput Phys 2018;363:158–77.